Notes: Duchin, Ozbas, and Sensoy (JFE, 2010)

Costly External Finance, Corporate Investment, and the Subprime Mortgage Credit Crisis

Contents

1 Big picture

- **Question.** How does a negative shock to the supply of external finance affect real corporate investment? The paper studies whether non-financial firms cut investment after the 2007 subprime mortgage credit crisis, and whether the cuts are larger when firms have less internal liquidity.
- **Setting.** The authors study U.S. publicly traded non-financial and non-utility firms using quarterly Compustat data. The main analysis compares the year before the crisis, July 1, 2006 to June 30, 2007, with the first year of the crisis, July 1, 2007 to June 30, 2008.
- **Core empirical idea.** The paper treats the early phase of the crisis as a credit-supply shock. It estimates differences-in-differences regressions that compare investment before and after the crisis as a function of cash reserves, short-term debt, financial constraints, and industry dependence on external finance.
- **Main result.** Corporate investment declines significantly after the crisis begins. In the base specification, quarterly capital expenditures fall by 0.109 percentage points of assets, or 6.4% of the unconditional mean.
- **Liquidity result.** The decline is much larger for firms with low cash reserves. With controls for firm fixed effects, Tobin's Q, and cash flow, a zero-cash firm experiences an estimated decline of 0.179% of assets, while higher pre-crisis cash substantially mitigates the decline.
- **Supply-shock interpretation.** The investment decline is greatest among firms that are financially constrained, have high net short-term debt, or operate in industries dependent on external finance. These are exactly the firms for which a contraction in external capital supply should bind most strongly.
- **Validation.** The authors show that similar patterns do not appear in placebo crises in 2003-2006 or after the September 11 demand shock. They also measure financial positions as far as four years before the crisis to reduce concerns that firms anticipated the shock.
- **Economic takeaway.** Cash holdings are not merely idle resources. In a credit crunch, internal liquidity allows firms to continue investment when external capital becomes costly or unavailable.

2 Incremental contribution of the paper

- **From financial crisis to real effects.** Much crisis research focuses on banks, mortgages, securitization, or lending standards. This paper moves from the financial sector to the real corporate sector by showing that the crisis reduced firm investment.
- **A firm-level test of credit supply.** The paper uses firm-level variation in pre-crisis liquidity, debt maturity, financial constraints, and industry external-finance dependence to distinguish supply-side credit constraints from broad demand weakness.
- **Predetermined liquidity as the key exposure.** Instead of using liquidity measured during the crisis, the paper measures cash and debt before the crisis. This makes the empirical design closer to an instrument-style strategy: pre-crisis financial positions predict exposure to the external-finance shock.

- **Multiple identification checks.** The paper combines several checks that point in the same direction: firm fixed effects, Q and cash-flow controls, four-year-lagged cash, placebo crises, the September 11 demand shock, financial-constraint splits, industry-level dependence measures, and short-term versus long-term debt comparisons.

- **Precautionary cash contribution.** The paper adds a "bright side" to the excess-cash literature. Cash that looks excessive in normal times may be valuable insurance against rare credit-market disruptions.

- **Policy and managerial relevance.** The results imply that credit-market disruptions can quickly transmit to corporate investment, and that corporate liquidity policy affects firms' ability to preserve real activity during financial crises.

3 Data and measurement

3.1 Sample construction

- The sample uses quarterly Compustat data for publicly traded firms, extracted from the April 30, 2009 update.

- The main sample period is July 1, 2006 to June 30, 2008: four quarters before and four quarters after the crisis date.

- The paper dates the crisis as beginning on July 1, 2007. This is conservative because many observers date the crisis to August 2007, but July 1 creates balanced pre- and post-crisis calendar quarters.

- The main analysis focuses on the first year of the crisis, when the crisis was primarily financial. A later section extends the sample through March 31, 2009, when demand-side effects became much stronger.

- Financial firms and utilities are excluded using SIC codes 6000-6999 and 4900-4949.

- The authors exclude very small firms and firms with extreme quarterly asset or sales growth, reducing noisy accounting observations and firms affected by mergers or major restructurings.

- The final main sample contains 26,421 firm-quarter observations for 3,668 firms.

Interpretation: The sample is designed to isolate normal non-financial corporate investment behavior around the credit shock, while avoiding industries and firms whose balance sheets or investment dynamics are structurally different.

3.2 Main investment and liquidity measures

- **Investment** is quarterly capital expenditures divided by total assets, expressed in percentage points.

- **Cash reserves** are cash and short-term investments divided by total assets. The main cash measure is taken once per firm, at the end of the last fiscal quarter ending before July 1, 2006.

- **Short-term debt** is debt in current liabilities divided by total assets.

- **Long-term debt** is long-term debt divided by total assets.

- **Net short-term debt** equals short-term debt minus cash, scaled by assets. This is a short-term liquidity-pressure measure.
- **Cash flow** is operating income before depreciation divided by total assets.
- **Tobin's Q** is computed using the Kaplan-Zingales-style market-value-to-book-value definition and bounded above at 10.

Interpretation: Cash and net short-term debt are the central exposure variables. Cash provides internal financing capacity, while short-term obligations create refinancing pressure precisely when credit markets tighten.

3.3 Financial constraints and industry dependence

- Financial constraints are measured using five standard proxies: the Kaplan-Zingales index, the Whited-Wu index, firm size, payout ratio, and bond ratings.
- For most measures, firms are classified as constrained or unconstrained by splitting at the median as of the last fiscal year ending before July 1, 2006.
- For bond ratings, firms with positive debt but no bond rating are treated as constrained; rated firms and zero-debt firms are treated as less constrained.
- Industry external-finance dependence and equity dependence are constructed following Rajan and Zingales. The authors compute historical industry medians over 2000-2005.
- Information asymmetry is proxied by industry-level dispersion in productivity growth, where productivity is sales per employee.

Interpretation: Financial-constraint variables capture firm-level exposure to external-finance costs. Industry dependence measures are useful because they are less directly chosen by the individual firm and therefore provide a separate source of identification.

3.4 Excess cash

- The paper estimates normal cash holdings using models based on Opler, Pinkowitz, Stulz, and Williamson (1999) and Dittmar and Mahrt-Smith (2007).
- Excess cash is the residual: actual cash minus predicted normal cash.
- The authors ask whether cash that looks excessive under normal models helps firms sustain investment during the crisis.

Interpretation: This test is important because standard excess-cash models may understate the value of liquidity in rare crisis states.

4 Models and empirical specifications

4.1 Base differences-in-differences specification

The central regression can be written as

$$Investment_{it} = \alpha_i + \beta After_t + \gamma(After_t \times Cash_{i,0}) + \delta Q_{it} + \phi CashFlow_{it} + \varepsilon_{it}. \qquad (1)$$

- $Investment_{it}$ is capital expenditures divided by assets for firm i in quarter t.
- $After_t$ equals one for quarters after July 1, 2007.
- $Cash_{i,0}$ is cash/assets measured at the end of the last fiscal quarter ending before July 1, 2006.
- Firm fixed effects α_i absorb time-invariant firm differences and the level effect of cash, since cash is measured once per firm.
- Tobin's Q and cash flow control for observable time-varying investment opportunities.
- Standard errors are heteroskedasticity-robust and clustered by firm, with a two-way clustered robustness check.

Interpretation: β is the post-crisis decline for a zero-cash firm. γ measures how much pre-crisis cash mitigates the investment decline.

4.2 Validation regressions

The paper repeats the base design with cash measured four years before the crisis and with placebo crisis dates:

$$Investment_{it} = \alpha_i + \beta After_t^{placebo} + \gamma(After_t^{placebo} \times Cash_{i,pre}) + Controls_{it} + \varepsilon_{it}. \tag{2}$$

- The four-year cash test asks whether the result depends on cash measured shortly before the crisis.
- Placebo crisis tests use July 1 of 2003, 2004, 2005, and 2006.
- The September 11 test asks whether a major negative demand shock generates a similar cash-investment pattern.

Interpretation: If cash simply proxies for future investment opportunities or demand sensitivity, similar results should appear in placebo periods or after 9/11. They do not.

4.3 Financial constraints and external-finance dependence

The base specification is estimated separately for constrained and unconstrained firms, and for firms in high- and low-dependence industries. The key comparisons are

$$\beta_C \quad \text{versus} \quad \beta_U, \qquad \gamma_C \quad \text{versus} \quad \gamma_U, \tag{3}$$

where C denotes constrained or dependent firms and U denotes unconstrained or non-dependent firms.

- A supply shock should reduce investment more for constrained or external-finance-dependent firms.
- Cash should matter more when external finance is more costly or less available.

Interpretation: These cross-sectional tests are the strongest evidence that the post-crisis investment decline is due to financing supply rather than a uniform fall in demand.

4.4 Leverage and debt maturity

The leverage tests replace cash with short-term debt, long-term debt, total debt, and net-debt measures:

$$Investment_{it} = \alpha_i + \beta After_t + \lambda(After_t \times Leverage_{i,0}) + \delta Q_{it} + \phi CashFlow_{it} + \varepsilon_{it}. \quad (4)$$

- Short-term debt should matter during a credit crunch because it must be refinanced soon.
- Long-term debt should matter less because it does not create the same immediate rollover pressure.

Interpretation: The paper finds the expected pattern: net short-term debt predicts larger investment cuts, while long-term debt does not have a similar effect.

4.5 Late-crisis specification

To study the later phase of the crisis, the paper estimates

$$\begin{aligned} Investment_{it} = \alpha_i &+ \beta_1 After_t + \gamma_1(After_t \times Cash_{i,0}) \\ &+ \beta_2 LateAfter_t + \gamma_2(LateAfter_t \times Cash_{i,0}) + Controls_{it} + \varepsilon_{it}. \end{aligned} \quad (5)$$

- $After_t$ covers July 1, 2007 to June 30, 2008.
- $LateAfter_t$ covers quarters after July 1, 2008 through March 31, 2009.
- The late period includes stronger demand-side effects, especially after the September-October 2008 market collapse.

Interpretation: Cash matters strongly in the early crisis period but not significantly in the late-crisis period. This is consistent with weaker investment demand and with firms using up their liquidity buffers.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies whether investment falls more after the crisis for firms that were more exposed to external-finance supply disruptions before the crisis.
- The design is not a randomized experiment. It is a differences-in-differences design with predetermined exposure variables and multiple validation checks.
- The strongest inference is about the differential effect of the credit-supply shock across firms with different liquidity and financing positions.

Interpretation: The paper is persuasive because several independent exposure variables all produce the same economic pattern.

5.2 Supply versus demand

- A pure demand shock should reduce investment broadly and should not necessarily be strongest among cash-poor, constrained, or external-finance-dependent firms.

- The early crisis period is chosen because the crisis was primarily a financial-sector shock before the broader real demand collapse intensified.
- The September 11 test is useful because 9/11 was a major negative demand shock, but it does not reproduce the paper's cash pattern.
- The late-after analysis shows that once demand effects become stronger, the special role of pre-crisis cash weakens.

Interpretation: The timing and cross-sectional evidence jointly support a financing-supply interpretation.

5.3 Endogeneity and measurement concerns

- Firms may choose cash because they anticipate future investment opportunities. The paper addresses this by using cash measured before the crisis, including cash measured four years earlier.
- Firm fixed effects control for time-invariant differences across firms.
- Q and cash flow control for observable investment opportunities, although they are imperfect proxies.
- Placebo crises check whether the cash-investment relation mechanically appears in normal periods.
- Industry external-finance dependence helps because it is less directly controlled by the individual firm.

Interpretation: The identifying assumption is not that cash is randomly assigned. The assumption is that pre-crisis cash is not positively correlated with unobserved within-firm investment-demand shocks exactly when the crisis begins. The validation tests make this assumption more plausible.

5.4 Limitations of the design

- The estimates are based on public firms and may not apply exactly to private firms.
- Q and cash flow may not fully capture changes in investment opportunities.
- The crisis date is approximate, although the authors choose a conservative and balanced date.
- The analysis of excess cash depends on estimated normal-cash models, which are noisy.
- The late-crisis results are harder to interpret because credit supply, demand, and depleted cash buffers all move at the same time.

Interpretation: The paper is strongest on the first-year supply effect. It is more cautious about the later crisis period, where demand and supply forces become harder to separate.

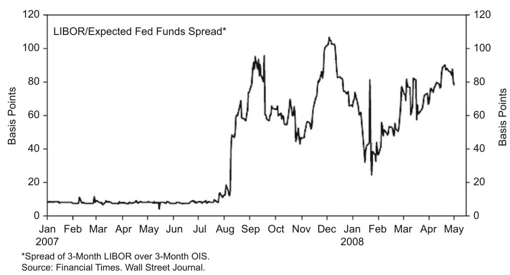
6 Tables and figures

6.1 Figure 1 and Figure 2: Credit-market shock and aggregate cash buffers

Read:

- Figure 1 plots the LIBOR-OIS spread. The spread is low and stable in early 2007, then jumps sharply around August 2007 and remains volatile through 2008.
- The spread is a useful visual measure of stress in short-term interbank funding markets and the rising cost of external finance.
- Figure 2 plots average cash/assets for non-financial firms from 1985 to 2008.
- Average cash holdings rise over much of the sample period but fall sharply by the end of the second quarter of 2008.

Economic message: The paper's empirical setting is a sudden increase in the cost and difficulty of external finance. Figure 2 shows why cash is central to the mechanism: firms entered the crisis with meaningful liquidity buffers, but those buffers were drawn down as the crisis unfolded.



ig. 1. London Interbank Offered Rate (LIBOR) – Overnight Indexed Swap Rate (OIS, Daily), as reported by Greenlaw, Hatzius, Kashyap, and Shin (2008).

(a) Figure 1: LIBOR-OIS spread during the onset of the crisis

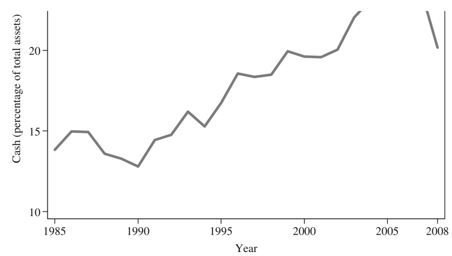


Fig. 2. Cross-sectional average cash (as a percentage of total assets) for non-financial firms, from 1985 to 2008 (*source*: Compustat).

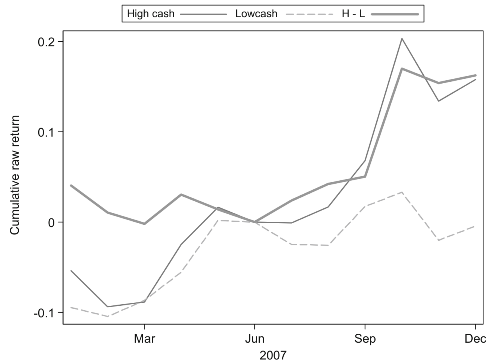
(b) Figure 2: Average cash/assets for non-financial firms

6.2 Figure 3: Cash-sorted stock returns

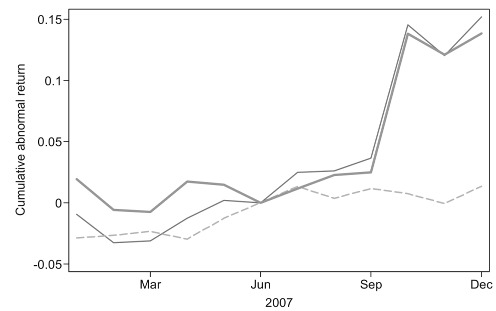
Read:

- Figure 3 sorts firms into high-cash and low-cash portfolios based on cash balances at the end of 2006.
- The raw-return panel shows that high- and low-cash portfolios move relatively similarly before the crisis, then diverge after the crisis begins.
- The abnormal-return panel shows a similar divergence after adjusting for Fama-French size and book-to-market styles.
- By the end of 2007, the cash-rich portfolio outperforms the cash-poor portfolio by roughly 15 percentage points in both raw and abnormal returns.

Economic message: The market appears to value cash during the crisis. This supports the idea that cash-rich firms' higher investment is not simply wasteful overinvestment; liquidity has a value-enhancing precautionary role when external finance tightens.



(a) Figure 3, raw returns



(b) Figure 3, abnormal returns

6.3 Table 1 and Table 2: Sample statistics and nonparametric investment patterns

Read:

- Table 1 reports the main sample summary statistics. Average quarterly capital expenditure is 1.695% of assets.
- Average pre-crisis cash/assets is 0.190, while average short-term debt/assets is 0.035 and long-term debt/assets is 0.169.
- The average quarterly cash-flow/assets ratio is 2.446%, and average Tobin's Q is 1.772.
- Table 2 sorts firms by pre-crisis cash, short-term debt, and net short-term debt.
- Investment falls significantly for low-cash firms, from 2.010% to 1.758% of assets per quarter, but is essentially flat for high-cash firms.
- Investment also falls most for firms with high net short-term debt, from 1.988% to 1.761% of assets per quarter.

Economic message: The raw data already show the paper's central mechanism. Firms with little liquidity or high short-term liquidity pressure cut investment much more after the crisis begins.

Table 1
Summary statistics.

This table reports summary statistics for the main sample of firm-year-quarter observations from July 1, 2006 to June 30, 2008. Cash is cash and short-term investments. Tobin's Q is the ratio of market value of assets to book value of assets following Kaplan and Zingales (1997), and is bounded above at 10. Cash flow is operating income before depreciation and amortization. Cash and debt variables are measured exactly once per firm, at the end of the last fiscal quarter ending before July 1, 2006.

	Mean	St. dev.	N Obs
Capital expenditure/assets (%)	1.695	2.301	26,421
Cash/assets	0.190	0.213	3,668
Short-term debt/assets	0.035	0.071	3,567
Long-term debt/assets	0.169	0.198	3,630
Cash flow/assets (%)	2.446	6.072	25,857
Tobin's Q	1.772	0.831	26,391
Market capitalization ($ millions)	5,313	20,813	26,505
Assets ($ millions)	5,121	23,418	27,129

(a) Table 1: Summary statistics

Table 2
Investment before and after the credit crisis.

This table presents difference-in-means estimates of firm-level quarterly investment (measured as the ratio of capital expenditures to total assets, in percentage points). Before crisis refers to the period July 1, 2006 to June 30, 2007. After crisis refers to the period July 1, 2007 to June 30, 2008. The reported means are cross-sectional averages of within-firm time-series averages for the relevant periods. To be included in the analysis, a firm must have capital expenditure data both before and after the crisis. Cash reserves is the ratio of cash and short-term investments to total assets at the end of the last fiscal quarter ending before July 1, 2006. ST debt is the ratio of short-term debt to total assets at the end of the last fiscal quarter ending before July 1, 2006. Net ST debt is the ratio of short-term debt minus cash to total assets at the end of the last fiscal quarter ending before July 1, 2006. Low, Medium, and High correspond to the first, second, and third terciles, respectively.

	Before crisis	After crisis	t-Statistic (difference)
Panel A: Cash reserves and average investment			
Low cash reserves	2.010	1.758	2.707
Medium cash reserves	1.875	1.795	0.937
High cash reserves	1.346	1.344	0.022
Panel B: Short-term debt and average investment			
Low ST debt	1.768	1.690	0.773
Medium ST debt	1.727	1.621	1.332
High ST debt	1.766	1.605	1.916
Panel C: Net short-term debt and average investment			
Low net ST debt	1.359	1.341	0.226
Medium net ST debt	1.915	1.815	1.123
High net ST debt	1.988	1.761	2.416

(b) Table 2: Investment before and after the credit crisis

6.4 Table 3 and Table 4: Base regressions, placebo crises, and the 9/11 demand shock

Read:

- Table 3 is the core regression table. Column 1 shows that investment falls by 0.109 percentage points of assets after the crisis begins.
- In the controlled specification, the estimated decline for a zero-cash firm is 0.179% of assets, and the interaction of After with cash reserves is 0.490 and statistically significant.
- The interpretation is that cash mitigates the post-crisis investment decline. A one-standard-deviation increase in cash reserves offsets a large share of the zero-cash decline.
- The result is robust to firm fixed effects, Tobin's Q, cash flow, industry-by-quarter fixed effects, and two-way clustered standard errors.
- Table 4 shows that the cash interaction remains positive when cash is measured four years before the crisis.
- Placebo crises in 2003-2006 do not reproduce the main positive cash effect. The 9/11 demand shock also does not produce the same pattern; the cash interaction is negative.

Economic message: Table 3 establishes the main investment-cash relation, and Table 4 makes the supply-shock interpretation more credible. The result is not a generic feature of cash-rich firms investing more after any date or after any adverse macro shock.

Table 3
Cash reserves and investment before and after the credit crisis.

This table presents estimates from panel regressions explaining firm-level quarterly investment for quarters with an end-date between July 1, 2006 and June 30, 2008. Investment is capital expenditures divided by total assets in percentage points. After is an indicator variable equal to one for fiscal quarters with an end-date after July 1, 2007, the approximate beginning of the credit crisis. Cash reserves is the ratio of cash to total assets at the end of the last fiscal quarter ending before July 1, 2006. *Q* is the ratio of market value of assets to book value of assets following Kaplan and Zingales (1997), and is bounded above at 10. Cash flow is operating income before depreciation and amortization divided by total assets in percentage points. All regressions include firm fixed effects. Specification 5 further includes industry-year-quarter fixed effects based on Fama-French 48-industry definitions. Standard errors (in brackets) are heteroskedasticity-consistent and clustered at the firm level, except for specification 6 which reports robust standard errors clustered by both firm and time (year-quarter) using the method of Petersen (2009). ***, **, or * indicates that the coefficient estimate is significant at the 1%, 5%, or 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
After	−0.109***	−0.185***	−0.172***	−0.179***		−0.179*
	[0.023]	[0.030]	[0.030]	[0.031]		[0.094]
After x Cash reserves		0.406***	0.476***	0.490***	0.481***	0.490***
		[0.103]	[0.105]	[0.109]	[0.110]	[0.185]
Q			0.202***	0.194***	0.180***	0.194***
			[0.046]	[0.049]	[0.050]	[0.059]
Cash flow				−0.022**	−0.023***	−0.022***
				[0.009]	[0.009]	[0.008]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.721	0.721	0.726	0.728	0.730	0.728
N Obs	26,421	26,382	25,842	24,937	24,797	24,937

(a) Table 3: Cash reserves and investment

Table 4
Cash reserves four years prior, placebo regressions, 9/11 negative demand shock.

This table presents several specifications for validation purposes. All variables are defined in Table 3. In column 1, Cash reserves is measured at the end of the last fiscal quarter ending before July 1, 2003 to explain firm-level quarterly investment for two years [−1,+1] around July 1, 2007, the approximate beginning of the credit crisis. Columns 2–5 report placebo regressions explaining firm-level quarterly investment for two years [−1,+1] around placebo crises occurring on July 1 of 2003, 2004, 2005, and 2006, respectively. In these placebo regressions, After is an indicator variable equal to one for fiscal quarters with an end-date after the placebo crisis, and Cash reserves is measured at the end of the last fiscal quarter ending one year before the placebo crisis. Column 6 reports a similar regression explaining firm-level quarterly investment for two years [−1,+1] around September 11, 2001, the events of which led to a negative demand shock. All regressions include firm fixed effects. Standard errors (in brackets) are heteroskedasticity-consistent and clustered at the firm level. ***, **, or * indicates that the coefficient estimate is significant at the 1%, 5%, or 10% level, respectively.

Specification:	Cash 2003 Q2	Placebo 2003	Placebo 2004	Placebo 2005	Placebo 2006	9/11 Demand
	(1)	(2)	(3)	(4)	(5)	(6)
After	−0.123***	−0.056**	0.102***	0.062**	−0.048*	−0.412***
	[0.032]	[0.023]	[0.022]	[0.026]	[0.027]	[0.028]
After x Cash reserves	0.246***	−0.154*	0.039	−0.234***	0.102	−0.287***
	[0.094]	[0.082]	[0.083]	[0.087]	[0.078]	[0.104]
Q	0.157***	0.216***	0.229***	0.323***	0.273***	0.193***
	[0.048]	[0.035]	[0.036]	[0.045]	[0.047]	[0.026]
Cash flow	−0.017*	−0.008	−0.009	−0.008	−0.007	−0.006
	[0.009]	[0.007]	[0.006]	[0.007]	[0.006]	[0.006]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.728	0.715	0.766	0.787	0.794	0.673
N Obs	21,142	21,719	21,436	23,406	23,546	21,637

(b) Table 4: Placebo crises and 9/11 demand shock

6.5 Table 5 and Table 6: Financial constraints and external-finance dependence

Read:

- Table 5 splits firms using five financial-constraint measures: Kaplan-Zingales index, Whited-Wu index, firm size, payout ratio, and bond ratings.
- Investment declines for both constrained and unconstrained firms, but the decline is consistently larger for constrained firms.
- In Panel B, the decline for zero-cash firms is also larger among constrained firms, and the cash interaction is usually larger for constrained firms.
- Table 6 uses industry-level external-finance dependence, equity dependence, and information asymmetry.
- Firms in industries historically dependent on external finance or external equity experience much larger investment declines.
- Cash is especially valuable in external-finance-dependent industries; for example, the After-by-cash interaction is 0.894 in the high external-finance-dependence group.

Economic message: These tables are central for identification. If the crisis is a financing-supply shock, the effect should be strongest where external finance is needed most or hardest to obtain. That is exactly what the estimates show.

Table 5

Financial constraints, cash reserves, and investment before and after the credit crisis.

This table presents estimates from panel regressions explaining firm-level quarterly investment for quarters with an end-date from July 1, 2006 to June 30, 2008. The regressions are estimated separately for subsamples of firms formed on the basis of financial constraints measured at the end of the latest fiscal year ending before July 1, 2006. For the first four measures of financial constraints (Kaplan-Zingales (1997) index, Whited-Wu (2006) index, firm assets, payout ratio), the subsamples comprise firms with financial constraint measures below and above the sample median. For bond ratings, the low subsample comprises unrated firms that have positive debt, and the high subsample comprises the rest (this includes firms with zero debt and no debt rating). All other variables are defined in previous tables. All regressions include firm fixed effects. Standard errors (in brackets) are heteroskedasticity-consistent and clustered by firm. ***, **, or * indicates that the coefficient estimate is significant at the 1%, 5%, or 10% level, respectively. *p*-Values are reported at the bottom of each panel for stated null and alternative hypotheses on the estimated coefficients *A* (*After*) and *AxC* (*After x Cash reserves*) for financially constrained (C) and unconstrained (U) subsamples.

Panel A: Change in investment for financially unconstrained and constrained firms										
	Kaplan-Zingales index		Whited-Wu index		Firm assets		Payout ratio		Bond ratings	
	Low	High	Low	High	Big	Small	High	Low	High	Low
After	−0.046**	−0.157***	−0.071***	−0.113***	−0.069***	−0.143***	−0.082***	−0.170***	−0.046*	−0.170***
	[0.022]	[0.041]	[0.024]	[0.037]	[0.024]	[0.040]	[0.032]	[0.039]	[0.028]	[0.038]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.617	0.711	0.685	0.744	0.739	0.711	0.645	0.754	0.735	0.707
N Obs	12,514	12,570	12,569	12,659	13,045	13,058	9,380	9,335	13,340	12,955
A H_o:C=U, H_a:C < U		0.009		0.172		0.057		0.041		0.004
Panel B: Change in investment for unconstrained and constrained firms conditional on cash reserves										
	Low	High	Low	High	Big	Small	High	Low	High	Low
After	−0.099***	−0.212***	−0.072**	−0.235***	−0.100***	−0.268***	−0.088*	−0.254***	−0.102***	−0.252***
	[0.037]	[0.049]	[0.035]	[0.055]	[0.035]	[0.059]	[0.048]	[0.056]	[0.039]	[0.050]
After x Cash reserves	0.283**	0.789*	0.129	0.607***	0.324*	0.635***	0.177	0.643***	0.345***	0.643***
	[0.112]	[0.419]	[0.187]	[0.142]	[0.169]	[0.145]	[0.196]	[0.183]	[0.133]	[0.178]
Q	0.119***	0.237**	0.229***	0.139***	0.139**	0.209***	0.187**	0.208***	0.105*	0.265***
	[0.038]	[0.097]	[0.068]	[0.053]	[0.055]	[0.059]	[0.079]	[0.067]	[0.060]	[0.072]
Cash flow	−0.002	−0.022*	−0.034	−0.006	−0.017*	−0.022**	−0.001	−0.024**	−0.016	−0.028***
	[0.005]	[0.012]	[0.033]	[0.006]	[0.010]	[0.010]	[0.013]	[0.012]	[0.012]	[0.011]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.629	0.716	0.701	0.749	0.747	0.718	0.663	0.756	0.735	0.719
N Obs	11,727	11,975	11,740	12,040	12,095	12,527	8,764	8,859	12,540	12,271
A H_o:C=U, H_a:C < U		0.035		0.007		0.007		0.012		0.008
AxC H_o:C=U, H_a:C > U		0.122		0.020		0.082		0.042		0.090

(a) Table 5: Financial constraints and cash reserves

Table 6

External finance dependence, information asymmetry, and investment before and after the crisis.

This table presents estimates from panel regressions explaining firm-level quarterly investment for quarters with an end-date from July 1, 2006 to June 30, 2008. The regressions are estimated separately for subsamples of firms formed on the basis of industry-level measures of external-finance dependence, equity dependence, and information asymmetry estimated from 2000 to 2005. External-finance dependence is the industry-median proportion of investment not financed by cash flow from operations; Equity dependence is the industry-median ratio of equity to investment (following Rajan and Zingales, 1998). Information asymmetry is the industry standard deviation of productivity growth, as measured by the ratio of sales to the number of employees. For all measures, the low and high subsamples comprise firms with external-finance dependence and information asymmetry measures below and above the sample median, respectively. All other variables are defined in previous tables. All regressions include firm fixed effects. Standard errors (in brackets) are heteroskedasticity-consistent and clustered by firm. ***, **, or * indicates that the coefficient estimate is significant at the 1%, 5%, or 10% level, respectively. *p*-Values are reported at the bottom of each panel for stated null and alternative hypotheses on the estimated coefficients *A* (*After*) and *AxC* (*After x Cash reserves*) for external-finance dependent (D) and non-dependent (N) subsamples.

	External-finance dependence		Equity dependence		Information asymmetry	
Panel A: Change in investment and external finance dependence						
	Low	High	Low	High	Low	High
After	−0.005	−0.212***	−0.041	−0.169***	−0.098***	−0.126***
	[0.024]	[0.041]	[0.027]	[0.038]	[0.033]	[0.039]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.652	0.714	0.664	0.727	0.649	0.754
N Obs	13,073	12,905	12,483	13,495	11,811	11,755
A H_o:D=N, H_a:D < N		0.000		0.003		0.293
Panel B: Change in investment and external finance dependence conditional on cash reserves						
	Low	High	Low	High	Low	High
After	−0.010	−0.333***	−0.077**	−0.269***	−0.136***	−0.242***
	[0.029]	[0.054]	[0.035]	[0.052]	[0.046]	[0.055]
After x Cash reserves	0.134	0.894***	0.297**	0.704***	0.352*	0.612***
	[0.110]	[0.217]	[0.121]	[0.194]	[0.195]	[0.149]
Q	0.197***	0.193**	0.166***	0.246***	0.153**	0.191***
	[0.064]	[0.077]	[0.063]	[0.078]	[0.075]	[0.070]
Cash flow	−0.018*	−0.026*	−0.012	−0.035*	0.001	−0.027**
	[0.010]	[0.015]	[0.008]	[0.018]	[0.009]	[0.011]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.660	0.720	0.669	0.735	0.653	0.760
N Obs	12,258	12,255	11,762	12,751	11,019	11,138
A H_o:D=N, H_a:D < N		0.000		0.001		0.071
AxC H_o:N=D, H_a:N < D		0.001		0.038		0.144

(b) Table 6: External-finance dependence and information asymmetry

6.6 Table 7 and Table 8: Debt maturity and excess cash

Read:

- Table 7 studies leverage. Net short-term debt has a statistically significant negative relation with post-crisis investment changes.
- Long-term debt and total debt do not show the same significant pattern.
- This matters because short-term debt creates near-term refinancing needs, while long-term debt does not impose the same immediate liquidity pressure.
- Table 8 studies excess cash, defined as actual cash minus predicted normal cash based on standard cash-holding models.
- The interaction of After with excess cash is positive and significant under both the baseline and extended normal-cash specifications.

Economic message: Debt maturity and excess cash both reinforce the liquidity mechanism. Short-term obligations worsen the crisis effect, while even seemingly excess cash helps firms maintain investment.

Table 7
Leverage and investment before and after the credit crisis.

This table presents estimates from panel regressions explaining firm-level quarterly investment for quarters with an end-date from July 1, 2006 to June 30, 2008. Leverage is as of the last fiscal quarter ending before July 1, 2006, and is measured as short-term debt in column 1, long-term debt in column 2, total debt in column 3, net short-term debt (short-term debt minus cash) in column 4, net long-term debt (long-term debt minus cash) in column 5, and net debt (short- and long-term debt minus cash) in column 6, all normalized by total assets. All other variables are defined in previous tables. All regressions include firm fixed effects. Standard errors (in brackets) are heteroskedasticity-consistent and clustered at the firm level. ***, **, or * indicates that the coefficient estimate is significant at the 1%, 5%, or 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
After	−0.067**	−0.151***	−0.128***	−0.165***	−0.130***	−0.125***
	[0.028]	[0.036]	[0.037]	[0.031]	[0.026]	[0.027]
After x Leverage	−0.787	0.303	0.092	−0.612***	−0.006	−0.113
	[0.524]	[0.188]	[0.176]	[0.193]	[0.113]	[0.109]
Q	0.169***	0.180***	0.181***	0.218***	0.201***	0.220***
	[0.049]	[0.046]	[0.047]	[0.064]	[0.060]	[0.058]
Cash flow	−0.018**	−0.019**	−0.019**	−0.023**	−0.024**	−0.024**
	[0.008]	[0.008]	[0.008]	[0.011]	[0.011]	[0.011]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.729	0.741	0.741	0.730	0.742	0.744
N Obs	24,237	23,087	22,260	21,637	21,396	20,716
Leverage:	ST Debt	LT Debt	Debt	Net ST debt	Net LT debt	Net debt

(a) Table 7: Leverage and investment

Table 8
"Excess" cash and investment before and after the credit crisis.

This table presents estimates from panel regressions explaining firm-level quarterly investment for quarters with an end-date from July 1, 2006 to June 30, 2008. Excess cash is the residual cash to total assets at the end of the last fiscal quarter ending before July 1, 2006. Excess cash is defined relative to two models of optimal cash holdings, as presented in Opler et al. (1999) (in columns 1 and 2) and modified in Dittmar and Mahrt-Smith (2007) (in columns 3 and 4), estimated over the period from 1995 to 2004. All other variables are defined in previous tables. All regressions include firm fixed effects. Standard errors (in brackets) are heteroskedasticity-consistent and clustered at the firm level. ***, **, or * indicates that the coefficient estimate is significant at the 1%, 5%, or 10% level, respectively.

	Baseline specification		Extended specification	
	(1)	(2)	(3)	(4)
After	−0.103***	−0.078***	−0.096***	−0.071***
	[0.023]	[0.024]	[0.023]	[0.025]
After x Excess cash	0.679***	0.708***	0.822***	0.858***
	[0.118]	[0.124]	[0.125]	[0.130]
Q		0.182***		0.179***
		[0.049]		[0.049]
Cash flow		−0.021**		−0.021**
		[0.009]		[0.009]
Firm fixed effects	Yes	Yes	Yes	Yes
R^2	0.719	0.725	0.719	0.725
N Obs	25,619	24,309	25,619	24,309

(b) Table 8: Excess cash and investment

6.7 Table 9 and Table 10: Other spending and late-crisis period

Read:

- Table 9 repeats the main logic for other corporate spending measures: SG&A, R&D, net working capital excluding cash, and inventories.
- The interaction of After with cash reserves is positive and significant for all four spending categories.
- This suggests that liquidity affects a broad set of corporate spending margins, not only capital expenditures.
- Table 10 extends the sample through March 31, 2009 and separates the first crisis year from the late-crisis period.
- Cash continues to strongly mitigate investment declines in the first crisis year.
- In the late-after period, the cash interaction is positive but statistically insignificant, and the decline is largely explained by Q and cash flow.

Economic message: The early crisis looks like a binding credit-supply shock. Later in the crisis, demand weakness and depleted liquidity buffers become more important, so pre-crisis cash no longer has the same explanatory power.

Table 9
Other corporate spending before and after the credit crisis.

This table presents estimates from panel regressions explaining alternative firm-level quarterly spending measures for quarters with an end-date from July 1, 2006 to June 30, 2008. SG&A is sales, general, and administrative expense. R&D is research and development expense. NWC is net working capital excluding cash. Inventory is total inventories. All spending measures are divided by total assets and expressed in percentage points. All other variables are defined in previous tables. All regressions include firm fixed effects. Standard errors (in brackets) are heteroskedasticity-consistent and clustered at the firm level. ***, **, or * indicates that the coefficient estimate is significant at the 1%, 5%, or 10% level, respectively.

	SG&A	R&D	NWC	Inventory
After	−0.034	−0.084*	−0.584***	−0.102*
	[0.041]	[0.047]	[0.135]	[0.054]
After x Cash reserves	1.375***	0.715***	1.541***	1.167***
	[0.220]	[0.212]	[0.530]	[0.235]
Q	0.607***	0.418***	0.040	0.106
	[0.097]	[0.107]	[0.228]	[0.119]
Cash flow	−0.186***	−0.138***	0.135***	−0.033***
	[0.037]	[0.022]	[0.046]	[0.012]
Firm fixed effects	Yes	Yes	Yes	Yes
R^2	0.951	0.881	0.916	0.976
N Obs	23,244	11,913	24,253	24,098

(a) Table 9: Other corporate spending

Table 10
Cash reserves and investment before, after, and late-after the credit crisis.

This table presents estimates from panel regressions explaining firm-level quarterly investment for quarters with an end-date between July 1, 2006 and March 31, 2009. After is an indicator variable equal to one for fiscal quarters with an end-date between July 1, 2007 and June 30, 2008, the first year following the approximate beginning of the credit crisis. Late-after is an indicator variable equal to one for fiscal quarters with an end-date after July 1, 2008, which includes a negative shock to demand following the market meltdown in September–October 2008. Cash reserves is the ratio of cash to total assets at the end of the last fiscal quarter ending before July 1, 2006. All other variables are defined in previous tables. All regressions include firm fixed effects. Specification 5 further includes industry-year-quarter fixed effects based on Fama-French 48-industry definitions. Standard errors (in brackets) are heteroskedasticity-consistent and clustered at the firm level, except for specification 6 which reports robust standard errors clustered by both firm and time (year-quarter) using the method of Petersen (2009). ***, **, or * indicates that the coefficient estimate is significant at the 1%, 5%, or 10% level, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
After	−0.100***	−0.174***	−0.160***	−0.164***		−0.164*
	[0.023]	[0.029]	[0.030]	[0.031]		[0.093]
After x Cash reserves		0.396***	0.460***	0.473***	0.465***	0.473***
		[0.104]	[0.106]	[0.110]	[0.111]	[0.182]
Late-after	−0.110***	−0.109***	−0.043	−0.060		−0.060
	[0.029]	[0.038]	[0.041]	[0.040]		[0.074]
Late-after x Cash reserves		−0.012	0.128	0.159	0.159	0.159
		[0.142]	[0.138]	[0.133]	[0.134]	[0.169]
Q			0.197***	0.200***	0.189***	0.200***
			[0.050]	[0.051]	[0.053]	[0.056]
Cash flow				−0.021***	−0.022***	−0.021***
				[0.007]	[0.007]	[0.005]
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.703	0.704	0.708	0.712	0.713	0.712
N Obs	31,842	31,791	31,189	30,102	29,935	30,102

(b) Table 10: Early and late crisis periods

7 Short summary

- The paper studies how the 2007 subprime mortgage credit crisis affected corporate investment.
- The main empirical design compares investment before and after July 1, 2007 using firm-level differences in predetermined liquidity and financing exposure.
- Investment falls significantly after the crisis begins.
- The decline is larger for firms with low cash reserves and high net short-term debt.
- Cash reserves measured before the crisis mitigate the investment decline.
- Financially constrained firms experience larger investment cuts, and cash matters more for them.
- Firms in industries dependent on external finance or external equity experience larger declines.
- Placebo crises and the 9/11 demand shock do not generate the same cash-investment pattern.
- Seemingly excess cash helps firms sustain investment, highlighting a precautionary savings motive.
- In the later crisis period, demand-side forces become stronger and the role of pre-crisis cash weakens.

8 Conclusion

8.1 Core conclusion

Duchin, Ozbas, and Sensoy show that the subprime mortgage credit crisis had real effects on corporate investment. During the first year of the crisis, investment fell significantly, and the decline was greatest among firms most exposed to costly external finance: cash-poor firms, firms with high net short-term debt, financially constrained firms, and firms in industries dependent on external finance.

8.2 Economic intuition

The intuition is the standard financing-frictions mechanism. In a frictionless capital market, firms with profitable investment opportunities can always raise funds externally, so cash holdings should not determine investment. In a credit crunch, external finance becomes more expensive or unavailable. Firms with internal liquidity can continue funding projects, while firms without liquidity must cut investment even when projects remain valuable.

8.3 Incremental contribution revisited

The paper contributes by using the 2007 credit crisis as a large, plausibly external shock to the supply of finance for non-financial firms. It links the crisis to firm-level investment behavior and shows that the real effect depends strongly on firms' pre-crisis financial positions. It also contributes to the cash-holdings literature by showing that cash that appears excessive under normal models may have a valuable precautionary role during rare financial disruptions.

8.4 Implications for managers, investors, and policy

For managers, the paper suggests that liquidity policy should account for rare but severe funding shocks. For investors, it shows that cash can be value-enhancing when external capital markets become impaired. For policy, the results imply that shocks to financial institutions can transmit quickly to real investment through the corporate financing channel.

8.5 Limitations

The paper is observational and relies on the assumption that predetermined cash is not correlated with unobserved firm-specific investment-demand shocks after the crisis begins. Public-firm data may not fully represent private firms. Q and cash flow are imperfect controls for investment opportunities. The later-crisis period is harder to interpret because financing supply, investment demand, and firms' depleted cash buffers all move together.

8.6 Takeaway

The enduring takeaway is that corporate liquidity has real consequences. During normal times, cash may look idle or excessive. During a credit crisis, it becomes a financing substitute that helps firms preserve investment when external capital is costly.